STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 8: Mechanical Properties of Solids

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

Restoring force per unit area.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q2 1 mark

Pa.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q3 1 mark

DeltaL/L.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q4 1 mark

Within proportional limit, stress is proportional to strain.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q5 1 mark

Bulk modulus.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q6 1 mark

Yes.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q7 1 mark

Material is relatively stiff.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q8 1 mark

F/A.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q9 2 marks

5 x 10 Pa.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q10 2 marks

5 x 10-.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q11 2 marks

2 x 10 Pa.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q12 2 marks

0.01.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q13 2 marks

1 x 10- m.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q14 2 marks

Steel carries tensile stress effectively; concrete is strong in compression.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q15 3 marks

0.001 m.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q16 3 marks

2 x 10 Pa.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q17 3 marks

3 x 10 Pa.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q18 3 marks

DeltaL proportional to L.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q19 4 marks

Proportional limit ends linearity; elastic limit is the greatest stress for full recovery after unloading.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.

Q20 4 marks

DeltaL=FL/(AY)=3.75 x 10- m.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.